TWO-CIRCULANT BUTSON HADAMARD MATRICES OF ORDERS 46 AND 74

Abstract. A $BH(n,q)$ matrix is an $n \times n$ matrix with all entries in $\Omega_q = \{1, \zeta_q, \dots, \zeta_q^{q-1}\}$, $\zeta_q = e^{2\pi i/q}$, satisfying $HH^* = nI_n$. Szöllősi conjectured that a $BH(2p,6)$ matrix exists for every prime p; Czifra, Matolcsi and Szöllősi report that, hedging with "it seems", 46, 74, 86 and 94 are the even orders below 100 at which existence is still undecided. We exhibit a $BH(46,6)$ and a $BH(74,6)$, each in the two-circulant form $H = \left[\begin{smallmatrix} X & Y \\ Y^* & -X^* \end{smallmatrix}\right]$ of Czifra, Matolcsi and Szöllősi, whose method this is. Each matrix is specified by $2p$ integers printed below, and $HH^* = 2pI$ reduces to $p-1$ identities in $\mathbb{Z}[\zeta_6]$, so the objects are checkable by hand. Szöllősi's conjecture itself remains open.

1. Setting

Fix $q \geq 2$ and put $\zeta_q = e^{2\pi i/q}$, $\Omega_q = \{1, \zeta_q, \dots, \zeta_q^{q-1}\}$. A *Butson Hadamard matrix* $BH(n,q)$ is an $n \times n$ matrix H with every entry in Ω_q and $HH^* = nI_n$ [1]. Throughout, $q = 6$, we write $\zeta = \zeta_6$, and we use $\zeta^2 = \zeta - 1$, so that $\mathbb{Z}[\zeta] = \mathbb{Z} \oplus \mathbb{Z}\zeta$ is free of rank 2 and

$$|A + B\zeta|^2 = A^2 + AB + B^2 \qquad (A, B \in \mathbb{Z}), \tag{1}$$

a *Loeschian* number: a positive integer is Loeschian if and only if every prime $\equiv 2 \pmod 3$ occurs in it to an even power.

Szöllősi's thesis [5] states, verbatim and without side conditions:

For every prime p there is a $BH(2p,6)$ matrix.

It is cited as Conjecture 1.4.47 of that thesis in [2]. The existence table for $BH(n,6)$ with $n \leq 62$, $n \equiv 2 \pmod 4$, at [5, p. 32] carries exactly two question marks, at $n = 46$ and $n = 62$. Czifra, Matolcsi and Szöllősi [2] constructed a $BH(62,6)$; they also state, hedged with "it seems", that "$BH(46,6)$, $BH(74,6)$, $BH(86,6)$, and $BH(94,6)$ are the remaining undecided even orders less than 100", reading that off a comparison of the tables at [5, p. 32] and [4, p. 111]. We reproduce their hedge rather than harden it.

Theorem 1. *A $BH(46,6)$ matrix and a $BH(74,6)$ matrix exist. Explicit examples are $H(W_1)$, $H(W_2)$ and $H(W_3)$ of Section* 3.

Credit. The construction, the search method and the pruning identity used to find these matrices are all from [2]: their Section 4 introduces the

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two-circulant form (2) with first rows *simple of index* k for $k \mid p-1$, the meet-in-the-middle search over the autocorrelation vector, and the identity $|k+(p-1)\sum b_i|^2 = 2k^2p - |k+(p-1)\sum a_i|^2$, and applies them to $BH(62,6)$, $BH(82,6)$ and $BH(146,6)$, at $p = 31, 41, 73$. Whether the two orders below already occur in [2] we do not establish: its Table 1 lists the indices $k = 4, 6, 8$; none of these divides 22, but $k = 6$ divides 36, so $p = 37$ lies inside that range of indices.

2. The reduction to $p-1$ identities

For $e \in \mathbb{Z}_6^p$ write $x_i = \zeta^{e_i}$ and $X = \operatorname{circ}(x)$, meaning $X[r][c] = x_{(c-r) \bmod p}$; likewise Y from e_y. Set

$$H = \begin{bmatrix} X & Y \\ Y^* & -X^* \end{bmatrix} \in \mathbb{C}^{2p\times 2p}. \tag{2}$$

Every entry of H lies in Ω_6: this uses q even, since $-1 = \zeta^3$, so $-\overline{\zeta^a} = \zeta^{3-a}$. Explicitly, H has entries $\zeta^{E[i][j]}$ with

$$\begin{aligned} E[i][j] &= e_x[j-i], & E[i][p+j] &= e_y[j-i], \\ E[p+i][j] &= -e_y[i-j], & E[p+i][p+j] &= 3 - e_x[i-j], \end{aligned} \tag{3}$$

for $0 \le i, j < p$, with all index arithmetic taken mod p and all exponents mod 6. Put $\gamma_s(x) = \sum_{i\in\mathbb{Z}_p} x_i \overline{x_{i+s}} \in \mathbb{Z}[\zeta]$.

Lemma 2. *With X, Y circulant as above, $HH^* = 2p\, I_{2p}$ holds if and only if*

$$\gamma_s(x) + \gamma_s(y) = 0 \qquad \textit{for } s = 1, \dots, p-1. \tag{4}$$

Proof. From (2), $H^* = \left[\begin{smallmatrix} X^* & Y \\ Y^* & -X \end{smallmatrix}\right]$ and hence

$$HH^* = \begin{bmatrix} XX^* + YY^* & XY - YX \\ Y^*X^* - X^*Y^* & Y^*Y + X^*X \end{bmatrix}.$$

The conjugate transpose of a circulant is circulant: $X^* = \operatorname{circ}(w)$ with $w_m = \overline{x_{-m}}$. Circulants of the same order commute, so both off-diagonal blocks vanish identically, and $X^*X = XX^*$, $Y^*Y = YY^*$. Finally $(XX^*)[r][c] = \sum_t x_{t-r}\overline{x_{t-c}} = \gamma_{r-c}(x)$, so both diagonal blocks equal $\operatorname{circ}(\gamma_\bullet(x) + \gamma_\bullet(y))$. Since $\gamma_0(x) + \gamma_0(y) = p + p = 2p$ automatically, $HH^* = 2pI$ is exactly (4). □

So a candidate is decided by $p-1$ identities in a rank-2 lattice, each a sum of p roots of unity: about 500 integer additions at $p = 23$. All the arithmetic is exact, because collecting the exponent multiplicities $n_0, \dots, n_5$ of a sum $\sum_m n_m \zeta^m$ gives

$$\sum_{m=0}^{5} n_m\zeta^m = A + B\zeta, \qquad A = n_0 - n_2 - n_3 + n_5, \quad B = n_1 + n_2 - n_4 - n_5. \tag{5}$$

Two consequences of (4) organise the search. Summing (4) over $s = 1, \dots, p-1$ and adding γ_0 gives

$$\Big|\sum_i x_i\Big|^2 + \Big|\sum_i y_i\Big|^2 = 2p, \tag{6}$$

and by (1) both summands are Loeschian. Taking $y = x$ in (4) shows that a circulant $BH(p, 6)$ would force $|\sum_i x_i|^2 = p$; since 23 is prime and $\equiv 2 \pmod 3$ it is not Loeschian, so no circulant $BH(23, 6)$ exists. (This argument is special to $p = 23$: $37 = 3^2 + 3 \cdot 4 + 4^2$ *is* Loeschian.)

Following [2], for $k \mid p-1$ let $G_0 = \{h^k : h \in \mathbb{Z}_p^*\}$ be the subgroup of index k, and call $x \in \Omega_6^p$ *simple of index* k if $x_0 = 1$ and $i \mapsto x_i$ is constant on each of the k cosets of G_0 in $\mathbb{Z}_p^*$. Such an x is coded by k exponents $a_1, \dots, a_k \in \mathbb{Z}_6$, so a lane (p, k) holds 6^k vectors per side.

Lemma 3. *If x is simple of index k then $\gamma_s(x)$ depends only on the coset sG_0. If moreover $-1 \in G_0$ then $x_{-i} = x_i$, every $\gamma_s(x)$ is a rational integer, $\gamma_s = \gamma_{-s}$, and $2\sum_{s=1}^{(p-1)/2} \gamma_s(x) = |\sum_i x_i|^2 - p$.*

Proof. For $u \in G_0$ the map $i \mapsto ui$ permutes $\mathbb{Z}_p$, fixes 0 and preserves each coset of G_0, so it fixes x; replacing i by ui in $\gamma_s(x)$ gives $\gamma_{us}(x)$. If $-1 \in G_0$ then $x_{-i} = x_i$, whence $\gamma_{-s}(x) = \overline{\gamma_s(x)} = \gamma_s(x)$, so $\gamma_s(x)$ is real and lies in $\mathbb{Z}[\zeta] \cap \mathbb{R} = \mathbb{Z}$. The last identity is $\sum_{s\neq 0} \gamma_s = |\sum_i x_i|^2 - \gamma_0$ together with $\gamma_s = \gamma_{-s}$. $\square$

At $p = 23$ the admissible indices are $k \mid 22$, i.e. $k \in \{1, 2, 11, 22\}$; Proposition 4 rules out $k = 1$ and $k = 2$, and for $k = 11$ we have $G_0 = \{1, 22\} = \{\pm 1\}$, so Lemma 3 applies in full. By (6) the two norms are then odd (the coset sizes are 2, so $\sum_i x_i = 1 + 2\sum_{s=1}^{11} \zeta^{a_s}$ has A odd and B even), and the odd Loeschian numbers below 46 are $1, 3, 7, 9, 13, 19, 21, 25, 27, 31, 37, 39, 43$; hence the split (6) must be one of exactly

$$\{3, 43\}, \quad \{7, 39\}, \quad \{9, 37\}, \quad \{19, 27\}, \quad \{21, 25\}.$$

The two $BH(46, 6)$ matrices below realise two different members of this list.

Proposition 4. *No two-circulant matrix* (2) *with both first rows simple of index* 1 *or* 2 *is a $BH(46, 6)$, and none is a $BH(94, 6)$. Consequently the present method says nothing about $BH(94, 6)$: at $p = 47$ the admissible indices are $k \mid 46$, i.e. $k \in \{1, 2, 23, 46\}$, the first two are empty and the last two have 6^{23} and 6^{46} vectors per side.*

Proof. An index-1 vector is also simple of index 2, so it suffices to treat index 2: there $\sum_i x_i = 1 + \frac{p-1}{2}(\zeta^{a_1} + \zeta^{a_2})$, and $\zeta^{a_1} + \zeta^{a_2}$ is either 0 (when $a_2 = a_1 + 3$) or of modulus at least 1, since the six sums of two sixth roots of unity have moduli $2, \sqrt{3}, 1, 0$. Hence

$$\Big|\sum_i x_i\Big|^2 \in \{1\} \cup \Big[\big(\tfrac{p-1}{2} - 1\big)^2, \infty\Big), \tag{7}$$

which is $\{1\} \cup [100, \infty)$ at $p = 23$ and $\{1\} \cup [484, \infty)$ at $p = 47$. Now apply (6). If both norms are 1 then their sum is $2 \neq 2p$. If one is 1, the other must be $2p - 1$, i.e. 45 or 93, and neither lies in the set (7) (45 is not even Loeschian). If neither is 1, their sum is at least $200 > 46$, respectively $968 > 94$. The accompanying program checks the 36×36 ordered index-2 pairs at each prime directly against (4) and finds 0 solutions out of 1296 ordered (666 unordered) pairs at each of $p = 23, 47$. $\square$

3. The matrices

Each witness is the pair $(e_x, e_y) \in \mathbb{Z}_6^p \times \mathbb{Z}_6^p$ used in (2)–(3). For W_1 and W_3 we also print the coset code (k, g, G_0, a, b) from which (e_x, e_y) is expanded, g being a primitive root mod p and the j-th coset being $g^j G_0$; for W_2 only the expanded form is printed.

W_1: a $BH(46, 6)$. $p = 23$, $k = 11$, $g = 5$, $G_0 = \{1, 22\}$,

$$a = (1, 5, 5, 2, 4, 4, 3, 2, 2, 1, 0), \qquad b = (2, 1, 4, 3, 2, 1, 5, 3, 3, 0, 2),$$
$$e_x = (0, 1, 5, 4, 4, 5, 2, 2, 3, 0, 2, 1, 1, 2, 0, 3, 2, 2, 5, 4, 4, 5, 1),$$
$$e_y = (0, 2, 4, 1, 2, 1, 3, 3, 5, 2, 3, 0, 0, 3, 2, 5, 3, 3, 1, 2, 1, 4, 2),$$

with $|\sum x|^2 = 3$, $|\sum y|^2 = 43$, $3 + 43 = 46$.

W_2: a second $BH(46, 6)$, on a different Loeschian branch. $p = 23$, $k = 11$,

$$e_x = (0, 1, 2, 3, 4, 1, 4, 3, 4, 1, 4, 0, 0, 4, 1, 4, 3, 4, 1, 4, 3, 2, 1),$$
$$e_y = (0, 5, 3, 3, 5, 4, 1, 0, 1, 0, 4, 0, 0, 4, 0, 1, 0, 1, 4, 5, 3, 3, 5),$$

with $|\sum x|^2 = 9$, $|\sum y|^2 = 37$, $9 + 37 = 46$, and

$$\big(2\gamma_s(x)\big)_{s=1}^{11} = (-4, 2, -4, 6, -8, -2, 0, -16, 2, 6, 4) = -\big(2\gamma_s(y)\big)_{s=1}^{11},$$

whose entries sum to $-14 = 9 - 23$, as Lemma 3 requires.

W_3: a $BH(74, 6)$. $p = 37$, $k = 12$, $g = 2$, $G_0 = \{1, 10, 26\}$,

$$a = (1, 4, 2, 0, 4, 3, 4, 3, 2, 5, 5, 0), \qquad b = (1, 5, 4, 0, 0, 2, 1, 2, 4, 2, 0, 3),$$
$$e_x = (0, 1, 4, 2, 2, 0, 0, 2, 0, 4, 1, 4, 4, 0, 5, 4, 4, 3, 3,$$
$$0, 4, 5, 3, 0, 3, 5, 1, 4, 5, 5, 2, 5, 3, 2, 2, 3, 4),$$
$$e_y = (0, 1, 5, 4, 4, 3, 0, 4, 0, 0, 1, 1, 0, 3, 2, 5, 0, 2, 2,$$
$$3, 5, 0, 2, 0, 2, 0, 1, 1, 0, 2, 4, 2, 2, 4, 4, 2, 1),$$

with $|\sum x|^2 = 31$, $|\sum y|^2 = 43$, $31 + 43 = 74$. Here $|G_0| = 3$ and $-1 \notin G_0$, so only the first half of Lemma 3 applies and the γ_s are genuinely non-real: $\gamma_1(x) = -4 + \zeta$ and $\gamma_{14}(x) = 6 - 7\zeta$.

Proof of Theorem 1. Each e above has length p, first entry 0 and all entries in $\{0, \ldots, 5\}$, so by (3) every entry of the corresponding H lies in Ω_6. By Lemma 2 it remains to check (4), which is 22 integer pairs for W_1 and W_2 and 36 for W_3; collecting exponent multiplicities and reducing with (5) gives 0 in both components in every case. The accompanying program performs that check and also, redundantly, forms all $\binom{47}{2} = 1081$ and $\binom{75}{2} = 2775$ inner products $\langle H_i, H_j\rangle$ for $i \leq j$ in exact $\mathbb{Z}[\zeta]$ arithmetic, obtaining $2p$ on the diagonal and 0 off it with no exceptions. $\square$

Remark. None of the following symmetries of the ansatz carries W_1 to W_2: swapping the two blocks, conjugating both first rows, and the multiplier action $x_i \mapsto x_{ui}$, $u \in \mathbb{Z}_{23}^*$. Each of the three preserves the unordered pair $\{|\sum x|^2, |\sum y|^2\}$, which is $\{3, 43\}$ for one witness and $\{9, 37\}$ for the other. We claim nothing beyond that: in particular we do not claim that $H(W_1)$ and $H(W_2)$ are inequivalent as Butson Hadamard matrices, whose equivalence group (row and column permutations and multiplications by sixth roots of unity) is far larger and does not preserve the two-circulant shape.

4. What is not settled

The conjecture remains open. It is universally quantified over infinitely many primes, and two further confirming instances cannot prove it. Nothing here refutes anything.

$BH(86, 6)$ **and** $BH(94, 6)$ **are still undecided.** We did not search $p = 43$. At $p = 47$, Proposition 4 empties the two smallest admissible indices, and the remaining two, $k = 23$ and $k = 46$, were not searched; so the method as run is silent at $n = 94$. Neither order is excluded by anything here.

$n = 46$ **is the smallest undecided *even* order, not the smallest open case of** $BH(n, 6)$**.** The $BH(n, 6)$ existence table at [4, p. 111] marks $n = 31$, 37 and 43 undecided, all smaller than 46.

No priority is claimed. Our literature search was incomplete — in particular the small-order existence table of [3] was not consulted — so we do not assert that these two orders were previously unrealised.

5. Reproduction

The objects are the $2p$ integers of Section 3; a reader who wants only Theorem 1 can build H from (3) and check (4) by hand. A short program, `verify.py`, does this from the printed vectors alone, in exact $\mathbb{Z}[\zeta]$ integer arithmetic with no floating point and no third-party package: it rebuilds each H, confirms all entries lie in Ω_6, verifies (4), forms every inner product $\langle H_i, H_j\rangle$ with $i \leq j$, re-derives the row-sum norms and their Loeschian splits, re-derives G_0 from $\{h^k \bmod p\}$ and checks that the coset codes expand to the printed e_x, e_y, recomputes the γ values quoted above, re-proves the enumeration of admissible splits at $2p = 46$, and carries out the exhaustive index-≤ 2 sweeps of Proposition 4. Its transcript accompanies this note. It does not re-derive the counts reported by the searches that produced W_1, W_2 and W_3, and nothing above depends on them.

References

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